

THE FIRST SPARK

SOUL MAP GENERATOR

Algorithm Specification & System Record

Source of truth: [soul_map_generator.py](#)

Companion systems: [webhook_server.py](#) · [monthly_regenerate.py](#) · [signatures.json](#)

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PROPRIETARY & CONFIDENTIAL

This document describes the complete computational algorithm used to generate Soul Maps. It is intended for internal filing, IP record, and engineering reference. Do not distribute outside authorized parties without written permission.

1. Purpose & Scope

The Soul Map Generator is a deterministic pipeline that transforms a person's **full legal (or preferred) name**, **date of birth**, and optionally **birth time + city** into a branded HTML Soul Map page. The map combines Pythagorean numerology, Western sun-sign astrology, Chinese zodiac, the proprietary Selector Model, Color Codex assignment, permanent identity minting (quantum signature / activation code / resonance frequency), narrative synthesis, and monthly-cycle updates.

Outputs are committed to the public GitHub Pages host (soul-maps.thefirstspark.shop) and optionally emailed to the purchaser. Core identity fields are permanent after first mint (first-write-wins registry).

2. End-to-End Pipeline

2.1 Buyer journey

- Sales page → Whop checkout (\$22 product).
- Post-purchase redirect → success.html intake form (name, DOB, time, city, email).
- POST /generate on webhook_server.py (background thread).
- generate_soul_map() builds HTML + summary dict.
- generate_monthly_update() builds current-month page.
- deploy_to_github() writes both files + updates archive index.
- add_subscriber() stores 12-month entitlement in subscribers.json.
- Resend email delivers live URL.
- Cron (1st of month): monthly_regenerate.py refreshes active subscribers.

2.2 Primary entry points

Function / Endpoint	Role
generate_soul_map(name, dob, time?, city?)	Full map HTML + summary dict
generate_monthly_update(name, dob, y?, m?)	Monthly cycle HTML
get_or_mint_signature(...)	Permanent UUID registry (first mint wins)
deploy_to_github(html, filename, ...)	GitHub Contents API commit
POST /generate	Webhook intake (async generation)
monthly_regenerate.py	Batch monthly updates for active subs

3. Inputs & Normalization

Field	Format	Required	Notes
full_name	string	Yes	NFKC normalize; collapse whitespace; strip ZWSP/BOM
birth_date	YYYY-MM-DD → date	Yes	Core numerology + sun + Chinese year
birth_time	HH:MM → (h, m)	No	Enables Moon/Rising + planets via kerykeion
birth_city	string	No*	Required with time for full chart

Field	Format	Required	Notes
birth_country	ISO-ish	No	Default US
memorial_date / birthday_from	string	No	Ceremony banner modes

Name canonicalization (normalize_name)

Unicode NFKC → replace exotic spaces with ASCII space → strip zero-width / BOM → collapse runs of whitespace → strip ends. Does **not** force Title Case (preserves McDonald-style casing). Letter math uppercases inside numerology only.

```
identity_key = casefold(normalize_name) + '|' + ISO date
```

4. Numerology Engine (Core Algorithm)

System: **Pythagorean** letter values. Master numbers preserved: **11, 22, 33** when preserve_masters=True.

4.1 Letter map (PYTHAGOREAN_MAP)

Value	Letters
1	A J S
2	B K T
3	C L U
4	D M V
5	E N W
6	F O X
7	G P Y
8	H Q Z
9	I R

Vowels for Soul Urge: A E I O U only (Y treated as consonant).

4.2 Reduction function

reduce_number(n, preserve_masters=True): while n > 9, if preserve_masters and n ∈ {11,22,33} return n; else n = sum of decimal digits.

4.3 Core numbers

Life Path

```
reduce( reduce(month) + reduce(day) + reduce(digit_sum(year)) ) - masters preserved on each step and final
```

Expression

```
name_to_number(all letters A-Z)
```

Soul Urge

```
name_to_number(vowels only)
```

Personality

```
name_to_number(consonants only)
```

Birthday

```
reduce(day of month)
```

Maturity

```
reduce(Expression + Life Path)
```

Hidden Passion

```
most frequent digit 1-9 in name letters; ties → lowest digit
```

Karmic Lessons

```
digits 1-9 absent from name letter values
```

Personal Year

```
reduce( reduce(month, no master) + reduce(day, no master) + reduce(digit_sum(calendar_year), no master), masters yes )
```

Personal Month

```
reduce(PersonalYear + calendar_month)
```

Personal Day

```
reduce(PersonalMonth + reduce(calendar_day, no master))
```

Pinnacles

```
P1=M+D; P2=D+Y; P3=P1+P2; P4=M+Y (M,D,Y pre-reduced without masters)
```

Challenges

```
C1=|M-D|; C2=|D-Y|; C3=|C1-C2|; C4=|M-Y| then reduce
```

4.4 Reference computation (audit fixture)

Name: **Katelin Jill Puzakulics** · DOB: **1988-05-19** · as of mid-2026

Field	Value
Life Path	5
Expression	11
Soul Urge	4
Personality	7
Birthday	1
Maturity	7
Hidden Passion	3
Karmic Lessons	4, 6
Personal Year 2026	7

Field	Value
Pinnacles	6 / 9 / 6 / 4
Challenges	4 / 7 / 3 / 3
Sun	Taurus
Chinese	Earth Dragon
Selector Layer	Temporal — Freedom frequency
Color Codex	Violet · The Transformer · Purple
Quantum (12-hex)	E1855DD782A3
Resonance Hz	824.43
Activation	1C245658[LP5]
Base filename	KJP51988.html

5. Astrology Layers

5.1 Sun sign (always)

Table-driven tropical date ranges (Aries Mar 21–Apr 19 ... Pisces Feb 19–Mar 20). Capricorn wraps year boundary (Dec 22–Jan 19). Implemented in `sun_sign()`.

5.2 Full chart (optional)

If `birth_time` and `birth_city` provided: `kerykeion.AstrologicalSubject` computes Moon, Rising (1st house), Mercury, Venus, Mars, Jupiter, Saturn. On failure, map still generates with a 'birth time needed' placeholder.

5.3 Chinese zodiac

```
animal = ANIMALS[(year - 4) mod 12]
```

```
element = ELEMENTS[(year - 4) mod 10] // Wood×2, Fire×2, Earth×2, Metal×2, Water×2
```

6. Selector Model & Color Codex

6.1 Selector layer (from Life Path)

LP	Layer	Frequency tag
1	Temporal	Pioneer — initiate timelines
2	Relational	Bridge — weave connection fields
3	Metaphysical	Expression — transmit through creation
4	Physics	Structure — build architecture
5	Temporal	Freedom — collapse fixed timelines
6	Relational	Harmony — calibrate collective fields
7	Metaphysical	Seeker — decode hidden patterns
8	Physics	Power — materialize abstract → concrete
9	Relational	Completion — integrate signal layers
11	Metaphysical	Master Intuitive
22	Physics	Master Builder
33	Relational	Master Teacher

6.2 Color Codex priority (first match wins)

1. Gold — LP/EX/SU is 22 or 33, OR ≥ 2 master numbers in core set
2. Violet — any single 11 on LP / EX / SU (Transformer)
3. Rose — Life Path 2 or 6 (Bond)
4. Silver — Life Path 9 (Mirror)
5. Spectrum by LP with EX/SU refinements (red/ember/yellow/cyan/green/blue)

7. Advanced / Proprietary Outputs

7.1 Quantum signature

SHA-256 of concatenated core numbers string `f"{LP}{EX}{SU}{PE}{BD}"`; first 12 hex chars uppercased. Display form: AAAA·BBBB·CCCC.

Note: same core numbers → same quantum (name not in hash). Identity uniqueness is name|DOB in the registry.

7.2 Soul resonance frequency (Hz)

base = 432.0 Hz. Each core number maps to a harmonic multiplier (1→1.0, 2→1.5, 3→1.25, 4→2.0, 5→1.667, 6→1.2, 7→1.875, 8→2.667, 9→1.111, 11→3.0, 22→4.0, 33→5.333). Weighted average of multipliers for [LP, EX, SU, PE, BD] × 432, rounded to 2 decimals.

7.3 Activation code

MD5 of `f"{NAME_UPPER}{DD}{MM}{LP}"` → first 8 hex upper + literal suffix `[LP{n}]`. Name is `normalize_name().upper()` so whitespace quirks do not mint divergent codes.

7.4 Permanent mint registry (signatures.json)

`get_or_mint_signature`: if `identity_key` already exists, reuse quantum / resonance / activation / codex fields (marketing claim of permanent UUID). Else compute, store locally, and best-effort push to GitHub via `GITHUB_PAT`. `formula_version` currently = 1.

7.5 Supporting modules

Module	Behavior
rarity_detection	Score 0–100 from master counts + pattern flags
predictive_windows	90-day scan: birthday-day month shifts, 11/22 master days, next PY
power_hour_calculation	Maps personal day → suggested local hour block
evolutionary_trajectory	Maps 9-year personal year cycle arc
oracle_mapping	LP/EX/SU → Major Arcana trinity
destiny_checkpoints	Pinnacle ages, master PYs ahead, Saturn returns ~29/58
shadow_integration_path	Practices for top karmic lessons
karmic_debt_remediation	Heuristic remedies (see audit notes)
auto_soul_synthesis	Template narrative when no hand-written NARRATIVES entry
auto_debugging_notes	Watch-list bullets from masters / density / lessons

8. Filenames & Deployment

Base ID: initials (first letter of each word) + month + year

Example: Aaron Joseph Thomas, 1988-09-24 → AJT91988.html

Monthly: {base}-{YYYY}{MM}.html e.g. KJP51988-202607.html

Deploy uses GitHub Contents API (create-or-update with SHA). Host: <https://soul-maps.thefirstspark.shop/{filename}>.
Archive index updated via update_index_html / _api_update_index.

9. Data Artifacts

Artifact	Purpose
signatures.json	Permanent mints: quantum, Hz, activation, codex, core nums
subscribers.json	Paid entitlement: email, dob, purchase, expiry (+365d), active
bonds.json	Family / resonance graph (related product)
HTML map pages	Customer-facing deliverable on GitHub Pages
NARRATIVES dict	Optional hand-authored synthesis overrides

10. Proprietary Claim Summary (for filing)

While individual Pythagorean and sun-sign methods are traditional, the following combination and extensions constitute the Soul Map product system of The First Spark:

- Multi-layer stack: numerology + tropical sun + Chinese element/animal + Selector Model layer mapping.
- Color Codex assignment tree keyed to birth numerology (not aesthetic random hash).
- Permanent identity mint (quantum hex + activation + Hz) with first-write-wins registry.
- Resonance frequency formula ($432 \text{ Hz} \times \text{averaged number harmonics}$).
- Auto narrative synthesis / debugging notes with master-density detection.
- Destiny checkpoints combining pinnacles, master-year scan, and Saturn returns in one table.
- Fulfillment automation: webhook → generate → GitHub Pages → Resend → 12-month monthly regen.
- Brand voice interpretation corpus (LIFE_PATH_MEANINGS and related dictionaries) as expressive IP.

11. Algorithm Audit Findings (2026-07-26)

This section is the engineering audit run against `soul-maps/soul_map_generator.py` and `webhook_server.py`. Severity: Critical / High / Medium / Low / Info.

11.1 Critical / High

H1 — Unauthenticated `/generate` endpoint

`POST /generate` has CORS allowlist for shop domains but no payment proof, shared secret, or Whop webhook signature. Anyone who can reach the server can queue free maps and enroll subscribers. Recommend: require Whop webhook or signed intake token; rate-limit by IP/email.

H2 — Master Personal Years unreachable

`personal_year()` pre-reduces month/day/year with `preserve_masters=False`, so intermediates never stay 11/22. Final sum max is $9+9+9=27 \rightarrow 9$. Master Personal Years (11/22/33) and `destiny_checkpoints` master-year scan can never fire. If product copy promises master years, change reduction policy (preserve masters on day/month before sum, or single-pass reduction).

H3 — `karmic_debt()` is a stub

Function always returns `{}`. True karmic debt (13/14/16/19 unreduced) is not computed. `karmic_debt_remediation()` instead flags ALL LP 4/5/7/1 as potential debt — false positives.

H4 — Filename collisions

`get_base_filename` uses initials+month+year only. John Doe and Jane Doe both 1990-05 $\rightarrow$ JD51990. Later deploy overwrites earlier map. Mitigate: include day, short hash of name, or quantum prefix.

11.2 Medium

M1 — Quantum signature not name-unique

Hash inputs are only core numbers. Two different people with identical LP/EX/SU/PE/BD share the same quantum hex. Registry key (name|dob) is unique, but marketing 'fingerprint' can collide.

M2 — Evolutionary trajectory + master years

cycle math uses `personal_year` as if always 1–9 ($\text{years_remaining} = 10 - \text{py}$). Broken if masters appear. HTML 'next transition' line uses a fragile birthday-offset expression.

M3 — Power hour minute field

`hour_mapping` values are (hour_a, hour_b) pairs but assigned to hour/minute $\rightarrow$ displays like 08:09 instead of a real clock range 08:00–09:00.

M4 — Y vowel policy undocumented in product

Y is always consonant (value 7). Traditional schools sometimes treat Y as vowel. Document this as intentional system rule on sales page to reduce disputes.

M5 — Ephemeral subscribers on ephemeral disk

`subscribers.json` on Railway/Render is not durable across deploys unless volume/Git sync. Risk of losing monthly entitlement list. Prefer GitHub-backed or DB store (same pattern as signatures).

M6 — Background thread fire-and-forget

Webhook returns 200 before generation completes. Client has no job id; failures only in server logs. Add status endpoint or queue with durable jobs.

11.3 Low / Content

L1 — PINNACLE_MEANINGS all say 'First pinnacle'

P2–P4 reuse first-pinnacle wording; destiny_checkpoints strips prefix but pinnacle section copy is off.

L2 — Capricorn sun wrap dead clause

Condition ($m == 12$ and $m > sm$) never true ($sm=12$). Capricorn still resolves via other branches — dead code only.

L3 — Hand narratives sparse

NARRATIVES only hardcodes a few names; everyone else gets auto_soul_synthesis (good fallback, less premium).

L4 — HTML injection surface

Name and narrative fields are substituted into HTML without HTML-escaping. Malicious name could inject script into GitHub Pages. Escape user-controlled strings before template insert.

11.4 What is solid

- Core Pythagorean math for LP/EX/SU/PE/BD is consistent and deterministic.
- normalize_name prevents silent identity drift from form whitespace/unicode.
- Master number preservation on Life Path path is correct for 11/22/33 intermediates.
- Sun sign cusps verified (e.g. Mar 20 Pisces / Mar 21 Aries; Dec 21 Sag / Dec 22 Cap).
- Chinese zodiac (year–4) mod tables match standard animal/element cycles.
- Color Codex priority tree is explicit and LP5+EX11 → violet (Transformer) matches fixture.
- Signature first-mint-wins correctly reuses permanent IDs across regenerations.
- Optional kerykeion degrades gracefully when time/city missing or library fails.

11.5 Recommended fix order

1. Lock /generate behind payment verification + rate limits.
2. Decide intentional master Personal Year policy; implement or remove master-year UI.
3. Implement real unreduced karmic debt (13/14/16/19) or rename 'potential' copy.
4. Disambiguate filenames (day + short name hash).
5. HTML-escape all user strings; persist subscribers like signatures.
6. Fix power-hour display as a range; expand hand narratives for VIP maps.

12. Change Control

Any change to number reduction rules, letter map, Color Codex priority, quantum/activation inputs, or identity_key format must bump SIGNATURE_FORMULA_VERSION and be recorded here. Existing mints must remain stable (first-write-wins) unless an explicit migration is approved.

Version	Date	Notes
1.0	2026-07-26	Initial filing copy + full algorithm audit

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